Course Name: Basics of Mechanical Engineering
Course Code: ME-101
Course Type: Institute Core

Contact Hours/Week: 3L **Course Credits:** 3

Course Objectives

- Gain fundamental knowledge of Thermodynamics, and I.C. Engines.
- Gain knowledge of steam formation and properties of steam.
- Develop skills for material selection for different devices/ components.
- Develop know how about SFD and BMD.
- Develop knowledge to solve simple problems on stress strain.

Unit Number	Course Content	Contact Hours
UNIT-01	**Thermodynamics:** Heat, Temperature, Specific heat capacity, Change of state, Path, Process, Cycle, Internal energy, Enthalpy, Zeroth law and First law, Second Law, Heat engine, Classification of heat engines, Steam formation, Enthalpy, Specific volume, Internal energy and dryness fraction of steam, use of steam tables, steam calorimeters.	07L
UNIT-02	**Analysis of thermodynamic cycles:** Efficiency of Carnot, Otto, Diesel cycle, Concept of heat addition, heat rejection, work and efficiency. Classification of IC Engines, Efficiencies of IC Engines, CRDI, MPFI, Concept of hybrid engines.	07L
UNIT-03	**Engineering Materials:** Types, properties of engineering materials, Use of materials under different environmental conditions, applications of Ferrous & Nonferrous metals, Ceramics, Polymers and alloys.	06L
UNIT-04	**Manufacturing Process:** Basic description of the manufacturing processes – Sand Casting, Forging, Rolling, Extrusion and their applications. Types of Metal Joining Processes, Arc Welding, Soldering, Brazing and their applications, Machine tools and basic machining operations such as Turning, Drilling, Milling and Grinding. Basics of CAD/CAM, Rapid and Additive manufacturing.	07L
UNIT-05	**Force and Structure Analysis:** Force and Equilibrium, Centre of Gravity-simple cases, Beams, SFD & BMD of cantilever and simply supported beams, Trusses-simple cases.	07L
UNIT-06	**Stress and Strain:** Fundamentals of Stress and strain, Thermal Strain and Stress, Poisson Ratio, Elastic Constants, Bending of Beams, Bending equation, Torsion equation, Torsion of Springs & shafts.	08L

Course Outcomes

Upon successful completion of the course, the students will be able to

CO1: To understand the basic concepts of thermodynamics and heat transfer system
CO2: To understand the principles of IC- engines
CO3: To understand the use of materials for various applications
CO4: To understand the function of machine tools and machining processes
CO5: To Understand the basic concepts force, SFD and BMD
CO6: To understand the stress and strain, concept of bending and torsion

Books and References

1. J. Benjamin,Basic Mechanical Engineering,Pentex Books,9th Edition,2018.
2. R. K.Singal, M. Singal, and R. Singal, Basics of Mechanical Engineering, 2007.
3. G. S.Sawhney, Fundamentals of Mechanical Engineering, PHI.
4. P. K.Nag, Engineering Thermodynamics, 2018.
5. B.K.Agrawal, Introduction to Engineering Materials, McGraw Hill Publication, 2008.
6. S. Kalpakjian and S.R.Schmid, Manufacturing Processes & Engg Materials, 2022.